1. Suppose $f : X \to Y$ is a bijection. Show that if $A \subset X$, $(f^{-1})^{-1}(A) = A$. That is, show that the preimage of A under f^{-1} is A. Note that the two superscripts -1 are doing different work, so they do not simply cancel as they would if both meant "inverse function".

2. Let X be a topological space and let $A \subset B \subset X$. Show that the subspace topology on A as a subspace of B is the same as the subspace topology on A as a subspace of X. Note that B in this problem has the subspace topology (as a subspace of X).

3. Suppose X is a topological space and that $A \subset X$. Show that A is a compact subset of X if and only if A is a compact subset of A (with the subspace topology on A).

4. Let X and Y be topological spaces and let $A \subset X$ and $B \subset Y$. Then $A \times B$ is a subset of $X \times Y$ and has the subspace topology. But also, A has the subspace topology (as a subspace of X) and similarly B has the subspace topology, so $A \times B$ has the product topology as a product of two subspaces. Show that these two topologies are the same.

5. **(AI Free)** Suppose X and Y are topological spaces and Y is compact. Let $x \in Y$ and suppose W is an open set containing the fiber $\{x\} \times Y$. Show that there is an open set $V \subset X$ with $x \in V$ and $V \times Y \subseteq W$. Hint: cover the fiber with open rectangles, each contained in W. Show that you can, in fact, use finitely many of the rectangles. Then build V.

6. AI Use. Please describe specifically how you used AI to help complete this assignment.